

# VEDA AI ACADEMY

## EXAM ASSIGNMENT PAPER

Candidate Name: \_\_\_\_\_ Roll Number: \_\_\_\_\_

Class Section: \_\_\_\_\_ Date: \_\_\_\_\_

### General Instructions:

1. Read all questions carefully before answering.
2. Answer in the designated spaces or sheets clearly.

### Section A: Multiple Choice Questions

(Instruction: Choose the most appropriate answer for each question. Each question carries 1 mark.)

Q1. What is the primary goal of supervised learning in machine learning? [Marks: 1] (Easy)

- A. To predict an output based on labeled input data
- B. To discover hidden patterns in unlabeled data
- C. To reduce the dimensionality of data
- D. To identify anomalies in a dataset

Q2. Which of the following is a common step in the data preprocessing phase of a data science project? [Marks: 1] (Easy)

- A. Model deployment
- B. Feature scaling
- C. Hyperparameter tuning
- D. Report generation

Q3. In machine learning, what does 'overfitting' refer to? [Marks: 1] (Moderate)

- A. A model that performs well on training data but poorly on unseen data
- B. A model that is too simple to capture the underlying patterns in the data
- C. A model that takes an excessively long time to train
- D. A model that has too few features to make accurate predictions

Q4. Which type of Artificial Intelligence (AI) focuses on creating systems that can learn from data without explicit programming? [Marks: 1] (Easy)

- A. Rule-based AI
- B. Expert Systems
- C. Machine Learning
- D. Symbolic AI

### Section B: Short Questions

(Instruction: Answer the following questions briefly. Each question carries 2 marks.)

Q5. Briefly explain the difference between Artificial Intelligence (AI) and Machine Learning (ML). [Marks: 2] (Moderate)

Q6. List two common applications of Data Science in real-world scenarios. [Marks: 2] (Easy)

Q7. What is the purpose of a 'validation set' in machine learning model development? [Marks: 2] (Moderate)